

Chapter 1 Notes

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Fall 2025

1. **Definition 1.5.** Let S be a set. An *order* on S is a relation, denoted by $<$, with the following properties:

- (i) $\forall x, y \in S \ x < y, x = y, \text{ or } y < x.$
- (ii) $\forall x, y, z \in S \ x < y \text{ and } y < z \implies x < z.$

2. **Definition 1.7.** Let S be an ordered set and $E \subseteq S$. Then E is said to be *bounded above* and α is said to be an *upper bound* provided

$$(\forall x \in E)(\exists \alpha \in S) \ x \leq \alpha.$$

3. **Definition 1.8.** Let S be an ordered set, $E \subseteq S$, and E be bounded from above. Then $\alpha \in S$ is said to be the *supremum* of E and write $\alpha = \sup E$ provided

- (i) α is an upper bound for E ;
- (ii) If $\gamma < \alpha$, then γ is not an upper bound for E .

Similarly, $\beta \in S$ is said to be the *infimum* of E and write $\beta = \inf E$ provided

- (iii) β is a lower bound for E ;
- (iv) If $\gamma > \alpha$, then γ is not a lower bound for E .

4. **Definition 1.10.** Let S be an ordered set, $E \subseteq S$, $E \neq \emptyset$, and E be bounded from above. Then S is said to have the least-upper-bound property provided $\sup E \in S$.

5. **Theorem 1.11.** Let S be an ordered set with the least-upper-bound property, $B \subseteq S$, $B \neq \emptyset$, and B be bounded from below. Define L to be the set of all lower bounds of B . Then $\sup L = \inf B \in S$.

6. **Theorem 1.20.**

- (a) $\forall x, y \in \mathbb{R} \ x > 0 \implies \exists n \in \mathbb{Z}_{>0} \ nx > y.$
- (b) $\forall x, y \in \mathbb{R} \ x < y \implies \exists p \in \mathbb{Q} \ x < p < y.$

7. **Theorem 1.21.**

$$(\forall x \in \mathbb{R}_{>0})(\forall n \in \mathbb{Z}_{>0})(\exists! y \in \mathbb{R}_{>0}) \ y^n = x.$$